

Erwin Quanten

Staff Software Engineer – Java, Spring Boot, Microservices, AWS

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PROFESSIONAL SUMMARY

Staff Software Engineer specializing in **Java 8–21 backend engineering** for fintech, telecom, healthcare, logistics, and enterprise SaaS industries, with deep experience building **Spring Boot 2.x/3.x** microservices, REST APIs, event-driven systems, AWS cloud platforms, PostgreSQL, Kafka, Redis, Docker, Kubernetes, CI/CD, and secure distributed architectures; known for modernizing legacy Java and Spring applications, migrating monoliths into scalable services, resolving production performance and concurrency issues, improving API reliability by 18–27%, reducing deployment failures by 15–24%, and mentoring backend teams on clean architecture, observability, testing, and release quality.

SKILLS

- **Backend: Java 8–21, Spring Boot 2.x–3.x**, Spring MVC, Spring Security 5–6, Spring Data JPA, Hibernate 5.x–6.x, REST API, GraphQL, Microservices, OAuth2, JWT
- **Architecture: Event-Driven Architecture**, Domain-Driven Design, Clean Architecture, Hexagonal Architecture, Distributed Systems, Saga Pattern, CQRS, API Gateway
- **Cloud & DevOps: AWS**, EC2, ECS, EKS, Lambda, S3, RDS, CloudWatch, IAM, Docker 20+, Kubernetes 1.20+, Jenkins, GitHub Actions, GitLab CI
- **Databases: PostgreSQL 11–16**, MySQL 5.7–8.x, Oracle Database 11g–19c, MongoDB 4.x–6.x, Redis 5–7, Elasticsearch 7–8
- **Messaging: Apache Kafka 2.x–3.x**, RabbitMQ 3.x, AWS SQS, AWS SNS
- **Testing & Quality: JUnit 4/5**, Mockito, Testcontainers, WireMock, JaCoCo, SonarQube, Integration Testing, Contract Testing
- **Monitoring: Prometheus**, Grafana, ELK Stack, Splunk, OpenTelemetry, Spring Boot Actuator
- **Frontend Collaboration: React**, Angular, Vue, TypeScript, REST integration, API contract review

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

=Houston, TX

- Led **Java 17 and Spring Boot 3.x** backend modernization for enterprise SaaS platforms, replacing tightly coupled services with scalable microservices and improving release stability by **23%**.
- Architected secure **REST API** and event-driven modules using **Spring Security 6**, OAuth2, JWT, and Kafka 3.x, reducing authentication-related production incidents by **18%**.
- Migrated legacy **Spring Boot 2.3** services to **Spring Boot 3.x** and Jakarta EE namespaces, resolving dependency conflicts, deprecated APIs, and Hibernate compatibility issues.
- Improved database performance across **PostgreSQL 14–16** workloads by optimizing indexes, transaction boundaries, and JPA query plans, reducing slow API responses by **27%**.
- Resolved recurring memory leaks in high-traffic backend services by profiling JVM heap usage, tuning garbage collection, and correcting unclosed stream and cache references.
- Implemented resilient messaging flows with **Kafka 3.x**, retry topics, dead-letter queues, and idempotent consumers to prevent duplicate payment and billing events.
- Designed CI/CD pipelines with **GitHub Actions**, Docker 24, Kubernetes 1.27, and AWS EKS, reducing failed production deployments by **21%** through automated validations.
- Added observability with **OpenTelemetry**, Prometheus, Grafana, and Spring Boot Actuator, improving root-cause analysis speed for latency and timeout issues by **26%**.
- Solved concurrency bugs in order-processing workflows by applying optimistic locking, transaction isolation review, and distributed lock strategies using Redis 7.
- Implemented new customer onboarding features using **Java 17**, Spring Boot, PostgreSQL, and Kafka, keeping APIs backward compatible for existing enterprise clients.
- Mentored backend engineers on **clean architecture**, test coverage, API versioning, and production-readiness reviews, improving code quality scores by **19%** in SonarQube.
- Partnered with product, DevOps, and frontend teams to deliver secure backend services while maintaining flexible support for React and TypeScript client integrations.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Developed enterprise supply-chain and finance backend systems using **Java 8**, Spring Boot 1.5–2.0, Spring MVC, Hibernate 5, Oracle, and PostgreSQL.
- Migrated older Spring MVC modules into **Spring Boot 2.0** services, resolving XML configuration issues, bean initialization failures, and environment-specific startup errors.
- Improved batch-processing performance by **25%** by rewriting inefficient Hibernate queries, reducing N+1 selects, and tuning transaction boundaries for large datasets.
- Built secure internal **REST APIs** for inventory, procurement, and reporting workflows, improving data availability for operations teams by **19%**.
- Solved recurring production deadlocks by reviewing database locks, transaction ordering, and retry behavior, then applying safer update patterns in Java services.
- Implemented scheduled jobs and asynchronous workers for reconciliation workflows, reducing manual finance and warehouse corrections by **17%**.
- Added centralized logging, error tracking, and health checks to Spring services, making production support faster during nightly processing failures.
- Refactored legacy Java 7 code into cleaner Java 8 patterns using streams, optionals, service interfaces, and more maintainable exception handling.
- Worked closely with business analysts and infrastructure teams to deliver backend changes safely in regulated enterprise environments with strict release controls.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Delivered enterprise manufacturing and supply-chain applications using **Java 8**, Spring Boot 1.3-1.5, Spring MVC 4.x, Hibernate 4.x-5.x, Oracle Database 11g-12c, and early React 15-16 adoption.
- Introduced React components into selected internal dashboards where dynamic filtering and faster table interaction were needed, improving planner productivity by **18%** without forcing a full rewrite.
- Modernized older Spring MVC modules by separating controller logic, service logic, and repository access, reducing regression defects by **21%** during release testing cycles.
- Resolved a production issue where inventory updates were delayed by batch job failures, adding validation, retry handling, and better exception logging around Java scheduled processes.
- Implemented supplier tracking and shipment visibility features using Java REST endpoints, Oracle stored procedure integrations, and lightweight React views for operational teams.
- Supported migration from older Ant-based builds to Maven-based project structure, standardizing dependencies and making local setup easier for engineers across regions.
- Improved SQL performance by reviewing execution plans, adding indexes, and reducing unnecessary joins in high-volume reporting queries used by manufacturing planners.
- Collaborated with QA and business users to reproduce defects in order allocation screens, then delivered fixes that improved workflow accuracy by **16%** across recurring planning tasks.
- Balanced legacy platform constraints with newer frontend practices, using React only where it added clear value and keeping stable Java server-rendered flows where replacement was not justified.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Built and maintained internal enterprise applications using **Java 6–8**, Spring Framework 3–4, Hibernate 4, Oracle Database, JSP, and REST integrations.
- Upgraded selected Java 6 components to **Java 8**, resolving compiler issues, outdated dependencies, and incompatible application server configurations.
- Developed backend modules for procurement, inventory, reporting, and approval workflows, improving internal process turnaround time by **16%**.
- Resolved data inconsistency bugs by adding validation rules, database constraints, transactional service methods, and clearer exception handling in Java layers.
- Improved SQL performance by **21%** by rewriting joins, reducing unnecessary eager loading, and adding indexes for frequently used operational reports.
- Created reusable service components in Spring to reduce duplicate business logic across departments and improve long-term maintainability.

- Supported SOAP-to-REST integration transitions by building wrapper APIs while keeping older systems available during migration phases.
- Worked with QA teams to reproduce production defects, analyze logs, prepare fixes, and validate releases before deployment into business-critical environments.

Tessenderlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Developed Java-based internal tools using **Java 5–6**, Spring 2.5–3.x, Hibernate 3, Oracle Database, JSP, Servlet API, and Apache Tomcat.
- Fixed recurring form submission and validation bugs by improving server-side checks, exception handling, and database transaction consistency.
- Built reporting and workflow features for business users, reducing manual spreadsheet preparation by **18%** through automated Java backend processing.
- Supported migration from older servlet-based modules to Spring MVC patterns, improving code structure and reducing maintenance effort by **15%**.
- Collaborated with senior engineers to troubleshoot production errors, analyze stack traces, optimize SQL queries, and deliver stable backend enhancements.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009